Digitally Assisted Analog Front-End for Geophone Instrumentation

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***Abstract*—Programmable mixed-signal platforms provide an attractive foundation for scientific instrumentation by integrating configurable analog resources and embedded digital processing within a single device. However, the limited precision of their configurable analog building blocks often constrains front-end performance. This paper proposes a Digitally Assisted Analog Front-End (DA-AFE) that non-intrusively exploits the embedded mixed-signal resources of a programmable system-on-chip (PSoC) to improve the operating conditions of the configurable analog front end before signal acquisition while preserving a conventional analog acquisition path during normal operation.**

As a proof of feasibility, the proposed DA-AFE is demonstrated through an automatic foreground DC-offset calibration that sequentially re-centers the programmable-gain amplifier, band-pass filter, summing stage, and low-pass filter using the on-chip ADC, VDACs, programmable digital filter, and embedded processor.

The DA-AFE was experimentally evaluated on two independently assembled low-cost geophone acquisition boards under eight board–gain configurations. Relative to a common fixed-reference baseline, the calibrated low-pass output reduced the residual DC offset from 2.9–45.8% to 0.4–3.9% of the nominal ADC input span while using a single controller configuration across all reported conditions.

These results demonstrate that non-intrusive, embedded digital assistance provides a practical means of improving the operating conditions of configurable analog front ends without modifying the acquisition signal path. Although demonstrated through foreground DC-offset calibration, the proposed DA-AFE is not restricted to this function and can be extended to additional digitally assisted analog calibration and optimization tasks. Statistical validation over larger hardware populations and broader environmental characterization remain future work.

***Index Terms*—Digitally assisted analog design, analog front end, mixed-signal systems, geophone instrumentation, programmable system-on-chip (PSoC).**

I. Introduction

Modern programmable mixed-signal platforms have become an attractive foundation for low-cost scientific instrumentation because they integrate configurable analog resources together with analog-to-digital and digital-to-analog converters, programmable digital filters, direct memory access, and embedded processors within a single device. However, the configurable analog building blocks available in these platforms generally provide lower precision, larger offsets, and reduced matching than dedicated analog integrated circuits, limiting the achievable performance of the analog front end.

Rather than addressing these limitations through higher-grade analog components, manual trimming, or additional external circuitry, the proposed work follows the principles of digitally assisted analog design introduced by Murmann and Boser [1] and further developed by Murmann [2]. In this work, a *Digitally Assisted Analog Front-End* (DA-AFE) denotes a system in which embedded digital resources non-intrusively assist the analog circuitry to improve its effective operating conditions while leaving the acquisition signal path unchanged during normal operation. Although digitally assisted analog design has primarily been investigated at the integrated-circuit level, the present work extends the same philosophy to low-cost scientific instrumentation through a geophone acquisition platform.

Low-cost multichannel geophone acquisition systems provide a representative application for this DA-AFE. Surface-wave methods such as Multichannel Analysis of Surface Waves (MASW) estimate Rayleigh-wave dispersion to infer shear-wave velocity profiles for geotechnical site characterization [3], [4]. Low-cost embedded geophone interfaces have already been demonstrated [5], [6], and dedicated low-frequency conditioning circuits continue to be reported [7], but in multistage analog conditioning chains, accumulated amplifier and reference offsets progressively reduce the available dynamic range and, at high gain, may saturate intermediate stages before the low-amplitude seismic signal reaches the acquisition converter. Improving the effective operating conditions of the analog front end is therefore a precondition for exploiting the available acquisition range.

The proposed DA-AFE forms part of a low-cost reconfigurable Internet-of-Things (IoT) geophone acquisition platform [8] based on SM-24 moving-coil geophones [9]. The PSoC 5LP integrates programmable analog blocks, voltage DACs (VDACs), a delta-sigma ADC, programmable digital filters, DMA resources, and an Arm Cortex-M3 processor [10], providing all resources required to implement the proposed DA-AFE. As a proof of feasibility, this work implements an automatic foreground DC-offset calibration that sequentially recenters the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) before acquisition.

Previous offset-compensation techniques generally follow three approaches: numerical estimation and digital correction [11], [12], physical trimming of individual analog stages [13], or autozero and chopper stabilization techniques [14], [15]. In contrast, the proposed DA-AFE employs embedded digital resources only during foreground calibration, after which the analog signal chain operates conventionally throughout data acquisition.

The contributions of this work are threefold: (i) the proposal of a Digitally Assisted Analog Front-End (DA-AFE) that exploits embedded mixed-signal resources to improve the effective operating conditions of configurable analog hardware while preserving hardware simplicity; (ii) its demonstration through automatic foreground DC-offset calibration of a complete low-cost geophone conditioning chain; and (iii) an experimental evaluation on two independently assembled acquisition boards under multiple gain configurations using a single controller configuration across all reported conditions.

The remainder of this paper is organized as follows. Section II presents the proposed DA-AFE. Section III describes the foreground calibration method. Section IV reports the experimental validation, followed by the discussion and conclusion.

II. Proposed Digitally Assisted Analog Front-End

The proposed DA-AFE combines a conventional analog acquisition path with non-intrusive embedded digital assistance during foreground calibration. Unlike post-acquisition digital compensation, the embedded resources only establish the desired operating point of each analog stage; once calibration is completed they are disabled and acquisition proceeds entirely through the calibrated analog path.

The DA-AFE is implemented on a PSoC 5LP programmable mixed-signal platform. During foreground calibration, the embedded analog multiplexer, delta-sigma ADC, programmable digital filter, voltage DACs (VDACs), and embedded processor are non-intrusively repurposed as a mixed-signal feedback system that assists the analog circuitry while leaving the acquisition signal path unchanged. During normal operation, these resources resume their conventional acquisition functions.

The analog signal path, derived from the low-frequency geophone front end proposed by Ma *et al.* [16], is shown in Fig. 1. The conditioning chain consists of a programmable-gain instrumentation amplifier (PGA), a band-pass filter, an inverting summing stage, and a second-order low-pass filter. Four independently controlled VDACs shift the operating point of each stage during foreground calibration. Each reference node is decoupled by a 1 μF $\|$ 100 nF network that, together with the 16 kΩ internal output resistance of the VDAC, forms a single-pole low-pass filter at approximately 9 Hz, attenuating the 750 nV/$\sqrt{\mathrm{Hz}}$ VDAC noise density above that frequency [10].

The embedded digital resources sequentially measure the output of each analog stage through the internal analog multiplexer and delta-sigma ADC. The filtered DC estimate determines the VDAC correction required to recenter each stage. Calibration proceeds from the PGA to the low-pass filter because each stage inherits the residual offset propagated from the preceding stages.

Assuming quasi-static offsets during calibration, negligible loading between stages, and linear superposition of DC contributions, the residual offset at the output of stage i can be expressed as

$$y_i^{\mathrm{DC}} = \sum_{j<i} A_{ij} y_j^{\mathrm{DC}} + b_i + K_i u_i, \tag{1}$$

where A_{ij} represents the DC transfer gain from stage j to stage i, b_i denotes the intrinsic offset generated by stage i, and K_i is the DC gain relating the injected VDAC voltage u_i to the output of the corresponding stage.

This sequential operating principle is the key characteristic of the proposed DA-AFE: each stage is non-intrusively assisted until its operating point converges, after which the digital assistance is disabled and the analog front end operates as a conventional feed-forward signal chain throughout acquisition.

III. Foreground Calibration Method

This section details the foreground calibration introduced in Section II: the workflow, the control law, and the parameter set used for every reported measurement.

A. Calibration Workflow

The calibration workflow is illustrated in Fig. 2. At boot, after a gain change, or on command, the firmware loads the stored DAC codes and verifies whether the previously established calibration remains valid. Since DC estimation assumes a stationary geophone and slowly varying internal states, a valid operating point immediately returns the system to acquisition. Otherwise, the PGA, BP, SUM, and LP stages are calibrated sequentially with filter reset and settling after each stage. A watchdog timer and sample-count limits bound the process; during acquisition, the controller remains inactive, the LP stage is selected, and the VDAC codes remain constant.

During calibration, the programmable digital filter operates as a 128-tap FIR DC estimator, whereas during acquisition it is reconfigured with coefficients matched to the geophone bandwidth. After calibration, the acquisition coefficients, filter state, and routing are restored. A gain-indexed EEPROM record stores the valid-stage mask, four DAC codes, version, and cyclic-redundancy check (CRC); failed calibrations retain their bounded best codes without overwriting a valid record. Fig. 3 shows the digital hardware implementing this workflow.

B. Control Law and Calibration Parameters

The same control framework is independently applied to each analog stage. For stage i, $\hat{y}_i[n]$ is the filtered signed ADC count, r_i is the desired reference count, and $e_i[n]$ is the residual. Since the calibration target is the reference level, $r_i = 0$ and therefore $e_i[n] = \hat{y}_i[n]$. The residual is first mapped to the 8-bit DAC domain:

$$\begin{aligned} \varepsilon_i[n] &= \mathrm{round}\left(e_i[n] \frac{q_{\mathrm{ADC}}}{q_{\mathrm{DAC}}} \right) \\ &= \mathrm{round}\left(\frac{e_i[n] V_{\mathrm{ADC,span}} N_{\mathrm{DAC}}}{N_{\mathrm{ADC}} V_{\mathrm{DAC,span}}} \right), \end{aligned} \tag{2}$$

where $q_x = V_{x,\mathrm{span}}/N_x$ is the quantization step and N_x the number of quantization levels. Nominal datasheet values are used throughout: $N_{\mathrm{ADC}} = 2^{18}$, $N_{\mathrm{DAC}} = 2^8$, $V_{\mathrm{ADC,span}} = 5000$ mV, and $V_{\mathrm{DAC,span}} = 4096$ mV, giving $q_{\mathrm{ADC}} = 5000/2^{18}$ mV/count and $q_{\mathrm{DAC}} = 16$ mV/code [10], [17].

Equation 2 produces an integer error in equivalent DAC codes. To avoid oscillation between adjacent codes when the ideal operating point is not realizable, the implementation adopts the conservative one-step floor

$$\Delta_{i,\min}^{\mathrm{impl}} = \max(1, \lceil |K_i| \rceil), \tag{3}$$

$$\bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \le \Delta_i, \\ \varepsilon_i[n], & \text{otherwise}, \end{cases} \tag{4}$$

Inside the deadband, Eq. 4 suppresses integration, while lock and refinement determine convergence. Table I lists the adopted values satisfying $\Delta_i \ge \Delta_{i,\min}^{\mathrm{impl}}$.

Under the assumptions of Section II, the DA-AFE requires a measurable stage output together with sufficient trim authority and range. The controller is implemented in positional proportional-integral (PI) form,

$$c_i^*[n] = c_{i,\mathrm{base}} - s_i \left[\frac{k_{\mathrm{P},i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{\mathrm{I},i}}{|K_i|} \sum_{q=0}^{n-1} \bar{\varepsilon}_i[q] \right], \tag{5}$$

where $c_{i,\mathrm{base}}$ is the baseline feedforward code and $s_i = \mathrm{sgn}(K_i)$. The integral term accumulates accepted residuals and resets inside the deadband. The VDAC-to-node gains $|K_i|$ were obtained by symbolic DC analysis of each stage in MATLAB, from the nominal component values of Fig. 1. For the two-op-amp instrumentation amplifier PGA [18] the resulting ratio is unity, so $K_{\mathrm{PGA}} = 1$.

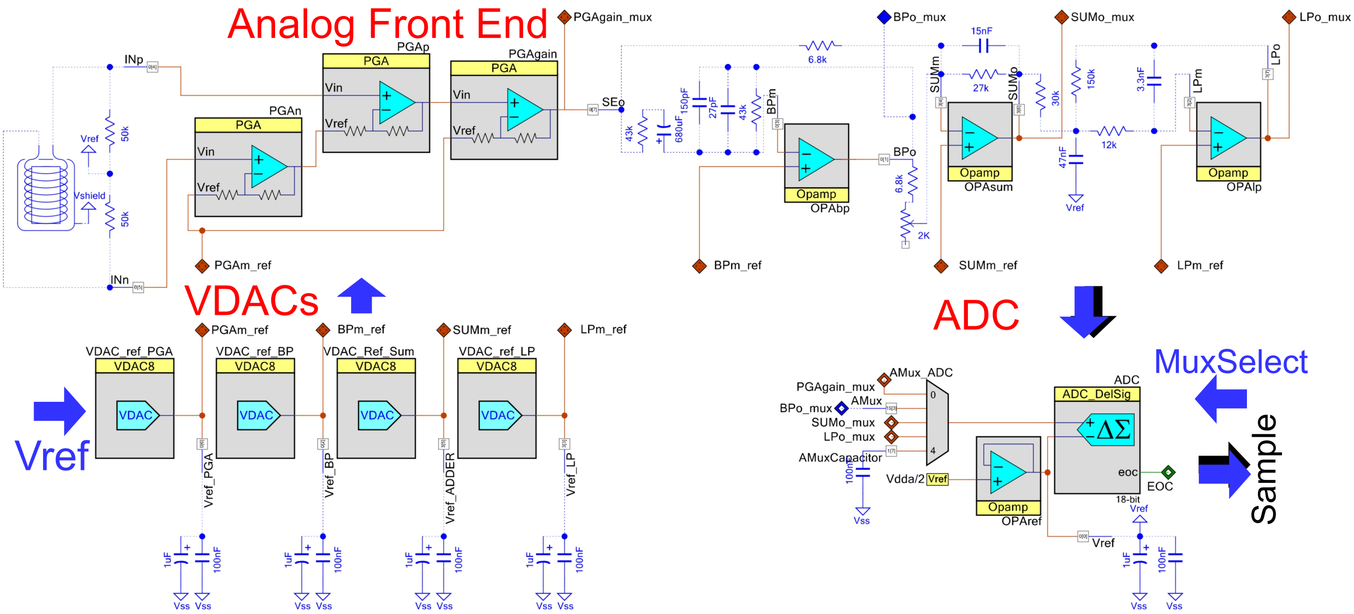

Fig. 1: Analog signal chain: four independent VDAC references trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

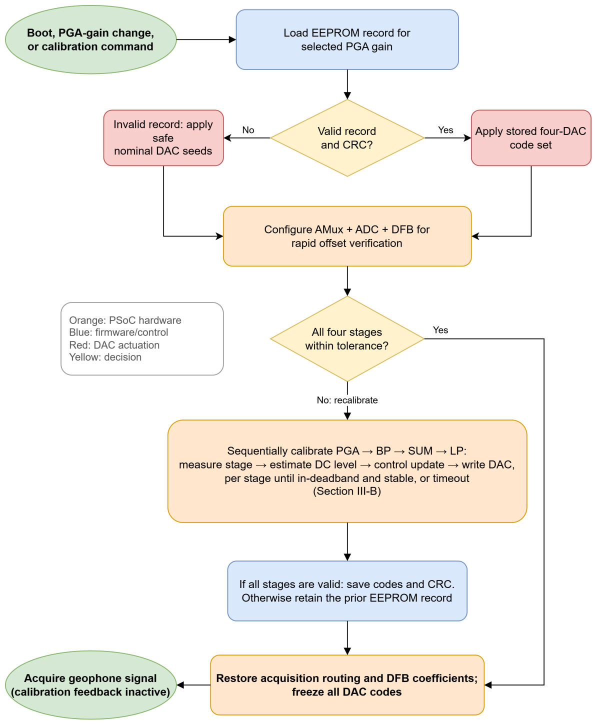

Fig. 2: Foreground calibration flow: stored-code verification, sequential trimming, and acquisition.

The command is clipped to the configured 0–255 DAC range. At either limit, integration is suspended whenever the residual would drive the command farther into saturation, preventing windup. Convergence requires a fixed number of consecutive in-deadband samples, while an adjacent-code trial is retained only if it reduces the filtered residual. At timeout, the best code is preserved and the stage is flagged.

The parameter set in Table I was selected empirically during preliminary bench tests and subsequently kept fixed for all reported measurements. No systematic optimization was performed, demonstrating robust operation with a single controller configuration across all evaluated boards and gain settings. Section V discusses the selected value of $\Delta_{\mathrm{PGA}} = 18$. Finer trim resolution could be achieved through current-output DACs or buffered injection topologies [17] without modifying the foreground calibration method.

TABLE I: Parameters used for the reported measurements. Deadbands are in equivalent DAC codes; S/L/R are settle/lock/refine samples at 2604 Hz, whereas Timeout is in seconds.

Stage	$\lvert K_i \rvert$	$\Delta_{i,\min}^{\mathrm{impl}}$	Δ_i	$k_{\mathrm{P},i}$	$k_{\mathrm{I},i}$	S/L/R	Timeout (s)
PGA	1	1	18	10^{-3}	3×10^{-4}	512/512/1024	11.5
BP	1	1	3	10^{-4}	5×10^{-4}	512/1024/1024	17.3
SUM	7.89	8	10	10^{-4}	5×10^{-4}	512/512/1024	11.5
LP	6	6	8	10^{-4}	5×10^{-4}	1024/1024/2048	17.3

IV. Experimental Validation

The proposed DA-AFE was evaluated through foreground calibration on two independently assembled geophone acquisition boards under multiple gain configurations, followed by quantitative residual DC-offset measurements.

A. Prototype and Comparison Conditions

The experimental validation comprised eight board–gain configurations: all five PGA settings on Board A together with the $\times16$, $\times32$, and $\times50$ settings on Board B. Each configuration was evaluated through ten paired restarts ($N = 10$), representing repeated measurements of the same board and gain rather than independent devices.

A single fixed-reference baseline, $\mathbf{c}_{\mathrm{base}}$, selected empirically on Board B at $\times50$, was applied unchanged to all reported configurations. This shared baseline is deliberately not a per-configuration optimum, so the reported reductions upper-bound the benefit relative to fixed per-board trimming rather than replicating it. For each calibrated condition, the four analog stages were processed sequentially and the resulting DAC codes remained fixed during acquisition. Post-calibration EEPROM writes were disabled so that every restart began from the same baseline.

Board B at $\times1$ and $\times4$ were excluded because their baseline residuals already lay within the calibration deadbands, preventing corrective updates. A separate narrower-deadband diagnostic confirmed the expected behavior (Section V).

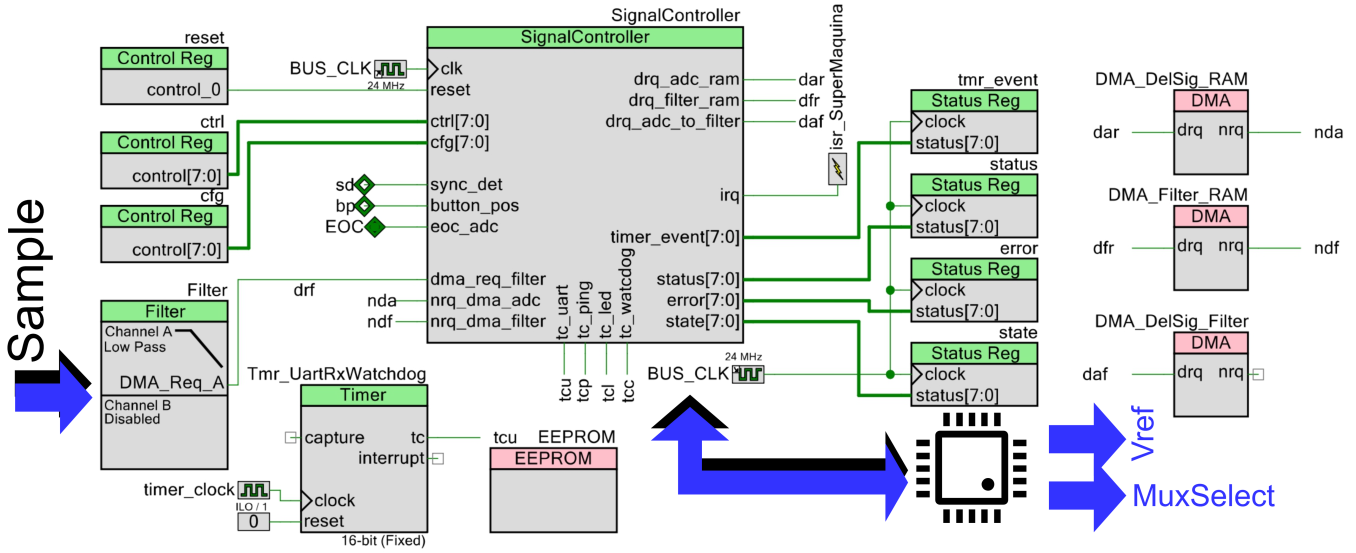

Fig. 3: Core PSoC digital hardware implementing the calibration workflow: signal controller, programmable digital filter, DMA channels, status registers, and EEPROM record.

B. Residual DC-Offset Measurement and Results

The LP output was selected as the primary performance metric because it directly feeds the acquisition ADC. Measurements at the PGA, BP, and SUM outputs document the sequential propagation of the foreground calibration through the analog signal chain.

A four-channel Tektronix TDS 2004B recorded the mean output voltage of the PGA, BP, SUM, and LP stages over a 1-s window after settling. Probe grounds were connected to $V_{\mathrm{ref}} = V_{\mathrm{DDA}}/2$, so each channel measured $V_{\mathrm{stage}} - V_{\mathrm{ref}}$. Within each restart, the fixed-reference condition was always recorded before the calibrated condition, preserving paired measurements.

Table II reports the mean and sample standard deviation of the normalized residual offset over ten paired restarts. Each column is one board–gain configuration and each row one analog stage in calibration order (PGA to LP), with the fixed-reference value stacked above the calibrated value in every cell, so a column read top-to-bottom shows the offset shrinking stage by stage. Markers $^{\dagger}/^{\ddagger}$ flag baselines $\geq 10\%$FS and the one calibrated mean above its baseline (Section V). Figure 4 shows representative calibration trajectories reconstructed by applying the 128-tap FIR used by the controller to oscilloscope records resampled at the controller rate.

Across the eight reported board–gain configurations, the proposed DA-AFE consistently reduced the LP residual DC offset without requiring board-specific or gain-specific controller retuning. Relative to the shared fixed-reference baseline, the residual offset decreased from 2.9–45.8%FS to 0.4–3.9%FS. Individual calibrated observations ranged from 0.1–5.2%FS (Table II). A one-sided sign test [19] on the paired LP measurements rejected the null hypothesis of no reduction in all eight configurations (10 of 10 paired restarts, exact $p \approx 10^{-3}$), a result that survives Bonferroni correction across the eight configurations. Stages whose baseline residual already lies inside their deadband are left unchanged by design, as in the BP row of Board B.

V. Discussion and Conclusion

The reported measurements show that embedded mixed-signal resources can recenter a complete low-cost conditioning

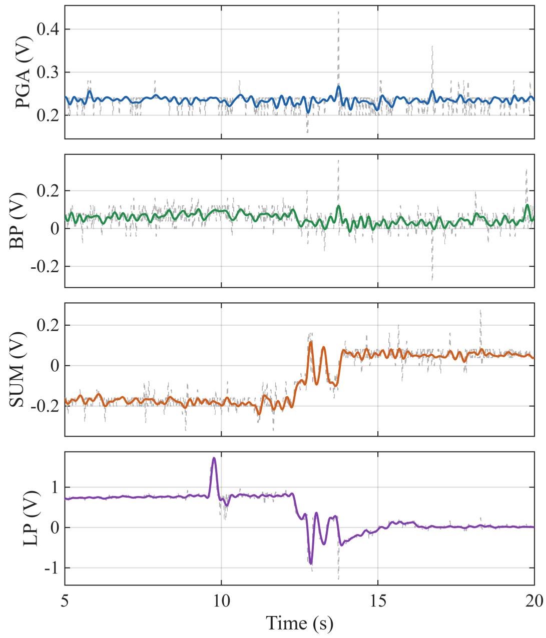

Fig. 4: Representative foreground calibration trajectories: stage deviations relative to V_{ref} during calibration at $\times 16$ gain (total differential gain $\times 32$). Dashed light-gray traces are oscilloscope measurements; colored curves are resampled 128-tap-FIR reconstructions, not internal logs (Section V).

chain without altering its acquisition path. From an instrumentation perspective, the proposed DA-AFE differs from two conventional approaches. Unlike precision analog front ends designed to operate close to the sensor's Johnson-noise limit [20], it trades analog component precision for embedded measurement and digital assistance. Likewise, unlike purely digital offset compensation, it physically recenters observable additive DC errors within the available VDAC authority and range instead of correcting them numerically after acquisition. The reported improvements therefore reflect the performance of the complete DA-AFE using a common fixed-reference baseline and a single controller configuration across all evaluated board and gain combinations rather than per-stage parameter optimization.

The proposed foreground calibration strategy also differs

TABLE II: Residual DC offset (%FS of the nominal 5-V ADC span), mean ± sample standard deviation over $N=10$ paired baseline-seeded restarts per reported board/gain cell. Within each cell, the upper and lower values correspond to the shared fixed-reference and calibrated conditions, respectively.

Stage	A, $\times 1$	A, $\times 4$	A, $\times 16$	A, $\times 32$	A, $\times 50$	B, $\times 16$	B, $\times 32$	B, $\times 50$
PGA	3.5 ± 0.2 3.4 ± 0.3	2.4 ± 0.1 2.1 ± 0.3	2.4 ± 0.1 1.9 ± 0.2	1.0 ± 0.2 0.5 ± 0.2	1.2 ± 0.3 1.0 ± 0.5	2.1 ± 0.1 1.7 ± 0.1	1.0 ± 0.2 0.5 ± 0.2	2.5 ± 0.1 1.8 ± 0.3
BP	1.5 ± 2.2 1.2 ± 0.2	4.9 ± 0.3 1.0 ± 0.2	3.8 ± 0.9 1.5 ± 1.5	7.0 ± 0.4 1.0 ± 0.5	1.0 ± 0.7 0.9 ± 0.5	0.66 ± 0.02 0.66 ± 0.02	0.66 ± 0.01 0.66 ± 0.01	0.65 ± 0.01 0.65 ± 0.02
SUM	$\mathbf{21.8 \pm 1.1}^{\dagger}$ 0.8 ± 0.2	6.7 ± 1.0 1.0 ± 0.3	4.9 ± 1.5 1.4 ± 1.5	7.4 ± 0.3 0.6 ± 0.2	0.4 ± 0.2 $\mathbf{0.8 \pm 0.6}^{\ddagger}$	4.5 ± 0.8 0.6 ± 0.5	8.0 ± 0.8 1.2 ± 0.3	0.9 ± 0.6 0.9 ± 0.6
LP	$\mathbf{45.8 \pm 3.1}^{\dagger}$ 3.2 ± 0.9	$\mathbf{42.6 \pm 4.2}^{\dagger}$ 3.9 ± 0.8	$\mathbf{29.6 \pm 6.8}^{\dagger}$ 2.3 ± 1.4	$\mathbf{12.4 \pm 1.3}^{\dagger}$ 1.7 ± 0.7	4.1 ± 2.1 1.2 ± 0.4	3.0 ± 0.1 0.7 ± 0.1	2.9 ± 0.1 0.4 ± 0.2	2.9 ± 0.1 0.7 ± 0.2

Lower is better. †Fixed reference $\geq 10\%$FS. ‡The marked calibrated mean is larger than its baseline; equality after rounding is not marked (Section V).

from continuous DC-servo architectures, which inherently introduce a settling-versus-bandwidth tradeoff through permanently closed feedback loops [15], [21]. Here, digital assistance is active only during foreground calibration. Once calibration is completed, the controller is disabled, the VDAC outputs remain constant, and the acquisition path contains no continuously active digital feedback.

The present implementation should be interpreted as a proof of feasibility rather than as an optimized analog front end. The relatively modest PGA improvement illustrates this point. Because $|K_{\mathrm{PGA}}| = 1$, the selected deadband ($\Delta_{\mathrm{PGA}} = 18$) accepts integer-error buckets through approximately ± 288 mV (continuous edge near ± 296 mV after rounding). Consequently, the 0.24-V residual shown in Fig. 4 reflects the selected stopping criterion rather than controller instability. A separate diagnostic experiment on Board B at $\times 4$ gain, performed outside the reported comparison, reduced the LP residual from a worst-case 2.98%FS before calibration to a best-case 0.84%FS after calibration with a narrower deadband, indicating that further improvement is possible through parameter optimization. Likewise, the only increase observed at the reported precision (Board A, SUM stage, $\times 50$ gain, from 0.4 to 0.8%FS) results from deadband acceptance and does not propagate to the downstream LP output.

The timing results also support the practical feasibility of foreground calibration. Visible correction in Fig. 4 spans approximately 7 s, while informal bench observations indicated completion in about 10 s. The configured timeout windows total 57.6 s (approximately 60.5 s including settling and refinement), excluding switching overhead; when a stored calibration is valid, only the per-stage settling intervals are executed (2560 samples, about 1.0 s at 2604 Hz). No timeout occurred during the reported experiments; the only timeout observed in bench testing was a board powered while its geophone was moving, whose AC transient violated the quasi-static assumption required during foreground calibration. Nevertheless, the present study was intentionally limited to two independently assembled boards and fixed calibration ordering; consequently, it does not establish population-level statistical performance, long-term drift behavior, inter-channel matching across an array, or optimal controller tuning. The reported evidence concerns the operating point of the conditioning chain rather than end-to-end seismic performance, although a fixed-reference offset such as the 21.8%FS measured at the SUM output already consumes more than a fifth of the amplifier and ADC input range before any seismic signal is applied.

Overall, the proposed DA-AFE offers a practical means of improving the operating conditions of low-cost configurable analog front ends by non-intrusively exploiting embedded mixed-signal resources. The reported foreground DC-offset calibration consistently reduced the LP residual offset across all evaluated boards and gain settings using a single controller configuration, while preserving a conventional analog acquisition path. Although demonstrated through DC-offset calibration, the DA-AFE is not restricted to this function and can be extended to other digitally assisted tasks in future work.

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